

3

L^p -Spaces

3.1 Inequalities

This section contains a number of useful inequalities.

Theorem 3.1.1: (*Markov's inequality*). Let f be a nonnegative measurable function on a measure space $(\Omega, \mathcal{F}, \mu)$. Then for any $0 < t < \infty$,

$$\mu(\{f \geq t\}) \leq \frac{\int f d\mu}{t}. \quad (1.1)$$

Proof: Since f is nonnegative, $\int f d\mu \geq \int_{\{f \geq t\}} f d\mu \geq t\mu(\{f \geq t\})$. $\square$

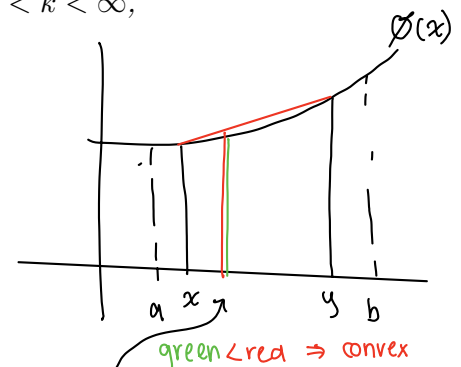
Corollary 3.1.2: Let X be a random variable on a probability space $(\Omega, \mathcal{F}, P)$. Then, for $r > 0$, $t > 0$,

$$P(|X| \geq t) \leq \frac{E|X|^r}{t^r}.$$

Proof: Since $\{|X| \geq t\} = \{|X|^r \geq t^r\}$ for all $t > 0$, $r > 0$, this follows from (1.1). $\square$

Corollary 3.1.3: (*Chebyshev's inequality*). Let X be a random variable with $EX^2 < \infty$, $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$. Then for any $0 < k < \infty$,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$



Proof: Follows from Corollary 3.1.2 with X replaced by $X - \mu$ and with $r = 2$, $t = k\sigma$. $\square$

Corollary 3.1.4: Let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be nondecreasing. Then for any random variable X and $0 < t < \infty$,

$$P(|X| \geq t) \leq \frac{E\phi(|X|)}{\phi(t)}.$$

Proof: Use (1.1) and the fact that $|X| \geq t \Rightarrow \phi(|X|) \geq \phi(t)$. $\square$

Corollary 3.1.5: (Cramer's inequality). For any random variable X and $t > 0$,

$$P(X \geq t) \leq \inf_{\theta > 0} \frac{E(e^{\theta X})}{e^{\theta t}}.$$

Proof: For $t > 0$, $\theta > 0$, $P(X \geq t) = P(e^{\theta X} \geq e^{\theta t}) \leq \frac{E(e^{\theta X})}{e^{\theta t}}$, by (1.1). $\square$

Definition 3.1.1: A function $\phi : (a, b) \rightarrow \mathbb{R}$ is called *convex* if for all $0 \leq \lambda \leq 1$, $a < x \leq y < b$,

$$\underbrace{\phi(\lambda x + (1 - \lambda)y)}_{E(X)} \leq \underbrace{\lambda\phi(x) + (1 - \lambda)\phi(y)}_{E\phi(X)}. \quad (1.2)$$

Geometrically, this means that for the graph of $y = \phi(x)$ on (a, b) , for each fixed $t \in (0, \infty)$, the chord over the interval $(x, x + t)$ turns in the counterclockwise direction as x increases.

More precisely, the following result holds.

Proposition 3.1.6: Let $\phi : (a, b) \rightarrow \mathbb{R}$. Then ϕ is convex iff for all $a < x_1 < x_2 < x_3 < b$,

$$\underbrace{\frac{\phi(x_2) - \phi(x_1)}{x_2 - x_1}}_{\text{slope}} \leq \frac{\phi(x_3) - \phi(x_2)}{x_3 - x_2}, \quad (1.3)$$

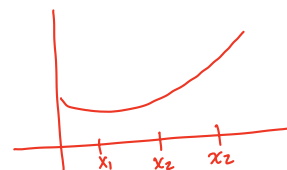
which is equivalent to

$$\frac{\phi(x_2) - \phi(x_1)}{x_2 - x_1} \leq \frac{\phi(x_3) - \phi(x_1)}{x_3 - x_1} \leq \frac{\phi(x_3) - \phi(x_2)}{x_3 - x_2}. \quad (1.4)$$

Proof: Let ϕ be convex and $a < x_1 < x_2 < x_3 < b$. Then one can write $x_2 = \lambda x_1 + (1 - \lambda)x_3$ with $\lambda = \frac{(x_3 - x_2)}{(x_3 - x_1)}$. So by (1.2),

$$\begin{aligned} \phi(x_2) &= \phi(\lambda x_1 + (1 - \lambda)x_3) \\ &\leq \lambda\phi(x_1) + (1 - \lambda)\phi(x_3) \\ &= \frac{(x_3 - x_2)}{(x_3 - x_1)}\phi(x_1) + \frac{(x_2 - x_1)}{(x_3 - x_1)}\phi(x_3) \end{aligned}$$

$$z = \lambda x + (1 - \lambda)y$$



READ optional

which is equivalent to (1.3). Also, since

$$\frac{\phi(x_3) - \phi(x_1)}{(x_3 - x_1)} = \lambda \frac{\phi(x_3) - \phi(x_2)}{(x_3 - x_2)} + (1 - \lambda) \frac{\phi(x_2) - \phi(x_1)}{(x_2 - x_1)},$$

(1.4) follows from (1.3).

Conversely, suppose (1.4) holds for all $a < x_1 < x_2 < x_3 < b$. Then given $a < x < y < b$ and $0 < \lambda < 1$, set $x_1 = x$, $x_2 = \lambda x + (1 - \lambda)y$, $x_3 = y$ and apply (1.4) to verify (1.2). $\square$

The following properties of a convex function are direct consequences of (1.3). The proof is left as an exercise (Problem 3.1).

Proposition 3.1.7: Let $\phi : (a, b) \rightarrow \mathbb{R}$ be convex. Then,

(i) For each $x \in (a, b)$,

$$\phi'_+(x) \equiv \lim_{y \downarrow x} \frac{\phi(y) - \phi(x)}{(y - x)}, \quad \phi'_-(x) \equiv \lim_{y \uparrow x} \frac{\phi(y) - \phi(x)}{(y - x)}$$

exist and are finite.

(ii) Further, $\phi'_-(\cdot) \leq \phi'_+(\cdot)$ and both are nondecreasing on (a, b) .

(iii) $\phi'(\cdot)$ exists except on the countable set of discontinuity points of ϕ'_+ and ϕ'_- .

(iv) For any $a < c < d < b$, ϕ is Lipschitz on $[c, d]$, i.e., there exists a constant $K < \infty$ such that

$$|\phi(x) - \phi(y)| \leq K|x - y| \quad \text{for all } c \leq x, y \leq d.$$

(v) For any $a < c, x < b$,

$$\phi(x) - \phi(c) \geq \phi'_+(c)(x - c) \quad \text{and} \quad \phi(x) - \phi(c) \geq \phi'_-(c)(x - c). \quad (1.5)$$

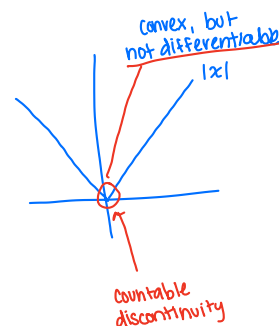
By the mean value theorem, a sufficient condition for (1.3) and hence, for the convexity of ϕ is that ϕ be differentiable on (a, b) and ϕ' be nondecreasing. A further sufficient condition for this is that ϕ be twice differentiable on (a, b) and ϕ'' be nonnegative. This is stated as

Proposition 3.1.8: Let ϕ be twice differentiable on (a, b) and ϕ'' be nonnegative on (a, b) . Then ϕ is convex on (a, b) .

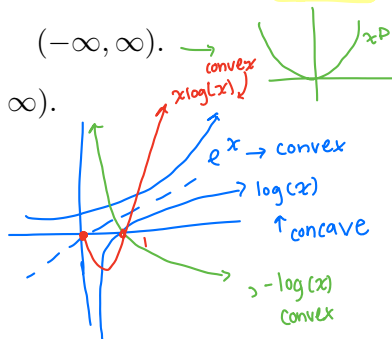
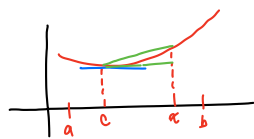
Example 3.1.1: The following functions are **convex** in the given intervals:

(a) $\phi(x) = |x|^p, p \geq 1, (-\infty, \infty)$.

(b) $\phi(x) = e^x, (-\infty, \infty)$.



$$\frac{\phi(x) - \phi(c)}{x - c} \geq \begin{cases} \phi'_+(c) \\ \phi'_-(c) \end{cases}$$



(c) $\phi(x) = -\log x, \quad (0, \infty).$

(d) $\phi(x) = x \log x, \quad (0, \infty).$

Remark 3.1.1: By Proposition 3.1.7 (iii), the convexity of ϕ implies that ϕ' exists except on a countable set. For example, the function $\phi(x) = |x|$ is convex on $\mathbb{R}$; it is differentiable at all $x \neq 0$. Similarly, it is easy to construct a piecewise linear convex function ϕ with a countable number of points where ϕ is not differentiable.

The following is an important inequality for convex functions.

Theorem 3.1.9: (*Jensen's inequality*). Let f be a measurable function on a probability space $(\Omega, \mathcal{F}, P)$ with $P(f \in (a, b)) = 1$ for some interval (a, b) , $-\infty \leq a < b \leq \infty$ and let $\phi : (a, b) \rightarrow \mathbb{R}$ be convex. Then

$$\phi\left(\int f dP\right) \leq \int \phi(f) dP, \quad (1.6)$$

provided $\int |f| dP < \infty$ and $\int |\phi(f)| dP < \infty$.

Remark 3.1.2: In terms of random variables, this says that for any random variable X on a probability space $(\Omega, \mathcal{F}, P)$ with $P(X \in (a, b)) = 1$ and for any function ϕ that is convex on (a, b) ,

$$\phi(EX) \leq E\phi(X), \quad \leftarrow \text{Random Variable} \quad (1.7)$$

provided $E|X| < \infty$ and $E|\phi(X)| < \infty$, where for any Borel measurable function h , $Eh(X) \equiv \int h(X)dP$.

Proof of Theorem 3.1.9: Let $c = \int f dP$. Applying (1.5), one gets

$$Y(\omega) \equiv \phi(f(\omega)) - \phi(c) - \phi'_+(c)(f(\omega) - c) \geq 0 \quad \text{a.e. } (P), \quad (1.8)$$

which, when integrated, yields $\int Y(\omega)P(d\omega) \geq 0$. Since $\int(f(\omega) - c)P(d\omega) = 0$, (1.6) follows. $\square$

Remark 3.1.3: Suppose that equality holds in (1.6). Then, it follows that $\int Y(\omega)P(d\omega) = 0$. By (1.8), this implies

$$\phi(f(\omega)) - \phi(c) = \phi'_+(c)(f(\omega) - c) \quad \text{a.e. } (P).$$

Thus, if ϕ is a strictly convex function (i.e., strict inequality holds in (1.2) for all $x, y \in (a, b)$ and $0 < \lambda < 1$), then equality holds in (1.6) iff $f(\omega) = c$ a.e. (P).

The following are easy consequences of Jensen's inequality (Problem 3.3).

Proposition 3.1.10: *Let $k \geq 1$ be an integer.*

consequence of
Jensen's Inequality
 $\Rightarrow$ for proof convert
expectations to sum

- (i) Let $a_1, a_2, \dots, a_k$ be real and $p_1, p_2, \dots, p_k$ be positive numbers such that $\sum_{i=1}^k p_i = 1$. Then

$$E[e^x] = \sum_{i=1}^k p_i \exp(a_i) \geq \exp\left(\sum_{i=1}^k p_i a_i\right) = e^{EX} \quad (1.9)$$

$\phi = e^x$
 $x = \{a_i : \text{w.p. } p_i\}$
 $EX = \sum a_i p_i$
 $E\phi(x) = \sum \phi(a_i) p_i$
 $= \sum e^{a_i} p_i$

- (ii) Let $b_1, b_2, \dots, b_k$ be nonnegative numbers and $p_1, p_2, \dots, p_k$ be as in (i). Then

$$\sum_{i=1}^k p_i b_i \geq \prod_{i=1}^k b_i^{p_i}, \quad (1.10)$$

$\phi(x) = -\log(x)$ $\leftarrow$ b/c $-\log(x)$ is convex
 $X = \begin{cases} b_1 & \text{w.p. } p_1 \\ b_2 & \text{w.p. } p_2 \\ \vdots \end{cases}$ and in particular,

$$\frac{1}{k} \sum_{i=1}^k b_i \geq \left(\prod_{i=1}^k b_i \right)^{\frac{1}{k}}, \quad (1.11)$$

$\leftarrow$ arithmetic mean $\leftarrow$ geometric mean

i.e., the arithmetic mean of $b_1, \dots, b_k$ is greater than or equal to the geometric mean of the b_i 's. Further, equality holds in (1.10) iff $b_1 = b_2 = \dots = b_k$.

$$\phi(x) = |x|^p$$

- (iii) For any a, b real and $1 \leq p < \infty$,

$$|a + b|^p \leq 2^{p-1}(|a|^p + |b|^p). \quad (1.12)$$

$$X = \begin{cases} a & \text{w.p. } p=0.5 \\ b & \text{w.p. } p=0.5 \end{cases}$$

$$\phi(EX) \quad E(\phi(X))$$

Inequality (1.10) is useful in establishing the following:

Theorem 3.1.11: (Hölder's inequality). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Let $1 < p < \infty$, $f \in L^p(\Omega, \mathcal{F}, \mu)$ and $g \in L^q(\Omega, \mathcal{F}, \mu)$ where $q = \frac{p}{p-1}$.

Then

$$\int |fg| d\mu \leq \left(\int |f|^p d\mu \right)^{\frac{1}{p}} \left(\int |g|^q d\mu \right)^{\frac{1}{q}}, \quad (1.13)$$

$$\text{i.e., } \|fg\|_1 \leq \|f\|_p \|g\|_q.$$

If $\|fg\|_1 \neq 0$, then equality holds in (1.13) iff $|f|^p = c|g|^q$ a.e. (μ) for some constant $c \in (0, \infty)$.

Proof: W.l.o.g. assume that $\int |f|^p d\mu > 0$ and $\int |g|^q d\mu > 0$. Fix $\omega \in \Omega$. Let $p_1 = \frac{1}{p}$, $p_2 = \frac{1}{q}$, $b_1 = c_1 |f(\omega)|^p$, and $b_2 = c_2 |g(\omega)|^q$, where $c_1 = (\int |f|^p d\mu)^{-1}$ and $c_2 = (\int |g|^q d\mu)^{-1}$. Then applying (1.10) with $k = 2$ yields

$$\frac{c_1}{p} |f(\omega)|^p + \frac{c_2}{q} |g(\omega)|^q \geq c_1^{\frac{1}{p}} c_2^{\frac{1}{q}} |f(\omega)g(\omega)|. \quad (1.14)$$

Integrating w.r.t. μ yields

$$1 \geq c_1^{\frac{1}{p}} c_2^{\frac{1}{q}} \int |f(\omega)g(\omega)| d\mu(\omega)$$

$$p_1 = \frac{1}{p} \quad q_2 = \frac{1}{q}$$

$$p_1 + p_2 = 1$$

which proves correlation ≤ 1
 extension of Cauchy-Schwarz inequality

$$\frac{1}{p} + \frac{1}{q} = 1$$

$$q = \frac{p}{p-1}$$

$$(p-1)q = p$$

$$p=2, q=2$$

$$p=1, q=\infty \leftarrow \text{assuming } \frac{1}{\infty} = 0$$

$$p=\infty, q=1$$

comes directly from (1.10)

which is equivalent to (1.13).

Next, equality in (1.13) implies equality in (1.14) a.e. (μ) . Since $1 < p < \infty$, by the last part of Proposition 3.1.10 (ii), this implies that $b_1 = b_2$ a.e. (μ) , i.e., $|f(\omega)|^p = c_2 c_1^{-1} |g(\omega)|^q$ a.e. (μ) . $\square$

Remark 3.1.4: (Hölder's inequality for $p = 1, q = \infty$). Let $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $g \in L^\infty(\Omega, \mathcal{F}, \mu)$. Then $|fg| \leq |f| \|g\|_\infty$ a.e. (μ) and hence,

$$\|fg\|_1 \equiv \int |fg| d\mu \leq \|f\|_1 \|g\|_\infty.$$

If equality holds in the above inequality, then $|f|(\|g\|_\infty - |g|) = 0$ a.e. (μ) and hence, $|g| = \|g\|_\infty$ on the set $\{|f| > 0\}$ a.e. (μ) .

The next two corollaries follow directly from Theorem 3.1.11. The proof is left as an exercise (Problem 3.9).

Corollary 3.1.12: (Cauchy-Schwarz inequality). Let $f, g \in L^2(\Omega, \mathcal{F}, \mu)$. Then

$$\int |fg| d\mu \leq \left(\int |f|^2 d\mu \right)^{\frac{1}{2}} \left(\int |g|^2 d\mu \right)^{\frac{1}{2}}, \quad (1.15)$$

$$\text{i.e., } \|fg\|_1 \leq \|f\|_2 \|g\|_2.$$

$$\frac{1}{p} + \frac{1}{q} = 1$$

$p=2, q=2 \rightarrow$

Corollary 3.1.13: Let $k \in \mathbb{N}$. Let $a_1, a_2, \dots, a_k, b_1, b_2, \dots, b_k$ be real numbers and $c_1, c_2, \dots, c_k$ be positive real numbers.

(i) Then, for any $1 < p < \infty$,

$$\sum_{i=1}^k |a_i b_i| c_i \leq \left(\sum_{i=1}^k |a_i|^p c_i \right)^{\frac{1}{p}} \left(\sum_{i=1}^k |b_i|^q c_i \right)^{\frac{1}{q}}, \quad (1.16)$$

$$\text{where } q = \frac{p-1}{p}.$$

(ii)

$$\sum_{i=1}^k |a_i b_i| c_i \leq \left(\sum_{i=1}^k |a_i|^2 c_i \right)^{\frac{1}{2}} \left(\sum_{i=1}^k |b_i|^2 c_i \right)^{\frac{1}{2}}. \quad (1.17)$$

$x = a_i$ w.p. c_i
 $y = b_i$ w.p. c_i
 ?
 check

Next, as an application of Hölder's inequality, one gets

Theorem 3.1.14: (Minkowski's inequality). Let $1 < p < \infty$ and $f, g \in L^p(\Omega, \mathcal{F}, \mu)$. Then

$\uparrow$
 p -th moment is finite

$$\left(\int |f+g|^p d\mu \right)^{\frac{1}{p}} \leq \left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}},$$

i.e., $\|f+g\|_p \leq \|f\|_p + \|g\|_p.$

$\triangle$ inequality in L^p

(1.18)

Proof: Let $h_1 = |f + g|$, $h_2 = |f + g|^{p-1}$. Then by (1.12),

$$|f + g|^p \leq 2^{p-1}(|f|^p + |g|^p),$$

implying that $h_1 \in L^p(\Omega, \mathcal{F}, \mu)$ and $h_2 \in L^q(\Omega, \mathcal{F}, \mu)$, where $q = \frac{p}{p-1}$.

Since $|f + g|^p = h_1 h_2 \leq |f| h_2 + |g| h_2$, by Hölder's inequality,

$$\int |f + g|^p d\mu \leq \left(\int |f|^p d\mu \right)^{\frac{1}{p}} \left(\int h_2^q d\mu \right)^{\frac{1}{q}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \left(\int h_2^q d\mu \right)^{\frac{1}{q}}. \quad (1.19)$$

But $\int h_2^q = \int |f + g|^p$ and so (1.19) yields (1.18). $\square$

Remark 3.1.5: Inequality (1.18) holds for both $p = 1$ and $p = \infty$.

Optional 3.2 L^p -Spaces

3.2.1 Basic properties

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Recall the definition of $L^p(\Omega, \mathcal{F}, \mu)$, $0 < p \leq \infty$, from Section 2.5 as the set of all measurable functions f on $(\Omega, \mathcal{F}, \mu)$ such that $\|f\|_p < \infty$, where for $0 < p < \infty$,

$$\|f\|_p = \left(\int |f|^p d\mu \right)^{\min\{\frac{1}{p}, 1\}}$$

and for $p = \infty$,

$$\|f\|_\infty \equiv \sup\{k : \mu(\{|f| > k\}) > 0\}$$

(called the *essential supremum* of f). In this section and elsewhere, $L^p(\mu)$ denotes $L^p(\Omega, \mathcal{F}, \mu)$. The following proposition shows that $L^p(\mu)$ is a vector space over $\mathbb{R}$.

Proposition 3.2.1: For $0 < p \leq \infty$,

$$f, g \in L^p(\mu), \quad a, b \in \mathbb{R} \quad \Rightarrow \quad af + bg \in L^p(\mu).$$

(2.1)

useful to show vector space

Proof:

CASE 1: $0 < p \leq 1$. For any two positive numbers x, y ,

$$\left(\frac{x}{x+y} \right)^p + \left(\frac{y}{x+y} \right)^p \geq \frac{x}{x+y} + \frac{y}{x+y} = 1.$$

Hence, for all $x, y \in (0, \infty)$

$$(x+y)^p \leq x^p + y^p. \quad (2.2)$$

It is easy to check that (2.2) continues to hold if $x, y \in [0, \infty)$. This yields $|af + bg|^p \leq |a|^p |f|^p + |b|^p |g|^p$, which, in turn, yields (2.1) by integration.

CASE 2: $1 < p < \infty$. By (1.12),

$$|af + bg|^p \leq 2^{p-1}(|af|^p + |bg|^p).$$

Integrating both sides of the above inequality yields (2.1).

CASE 3: $p = \infty$. By definition, there exist constants $K_1 < \infty$ and $K_2 < \infty$ such that

$$\mu(\{|f| > K_1\}) = 0 = \mu(\{|g| > K_2\}).$$

This implies that $\mu(\{|af + bg| > K\}) = 0$ for any $K > |a|K_1 + |b|K_2$. Hence, $af + bg \in L^\infty(\mu)$ and

$$\|af + bg\|_\infty \leq |a| \|f\|_\infty + |b| \|g\|_\infty.$$

□

Recall that a set $\mathbb{S}$ with a function $d : \mathbb{S} \times \mathbb{S} \rightarrow [0, \infty]$ is called a *metric space* if for all $x, y, z \in \mathbb{S}$,

- (i) $d(x, y) = d(y, x)$ (*symmetry*)
- (ii) $d(x, y) \leq d(x, z) + d(y, z)$ (*triangle inequality*)
- (iii) $d(x, y) = 0$ iff $x = y$

and the function $d(\cdot, \cdot)$ is called a *metric* on $\mathbb{S}$. Some examples are

- (a) $\mathbb{S} = \mathbb{R}^k$ with $d(x, y) = \left(\sum_{i=1}^k |x_i - y_i|^2\right)^{\frac{1}{2}}$;
- (b) $\mathbb{S} = C[0, 1]$, the space of all continuous functions on $[0, 1]$ with $d(f, g) = \sup\{|f(x) - g(x)| : 0 \leq x \leq 1\}$;
- (c) $\mathbb{S}$ is a nonempty set, and $d(x, y) = 1$ if $x \neq y$ and 0 if $x = y$. (This $d(\cdot, \cdot)$ is called the *discrete metric* on $\mathbb{S}$.)

The L^p -norm $\|\cdot\|_p$ can be used to introduce a distance notion in $L^p(\mu)$ for $0 < p \leq \infty$.

Definition 3.2.1: For $f, g \in L^p(\mu)$, $0 < p \leq \infty$, let

$$d_p(f, g) \equiv \|f - g\|_p. \quad (2.3)$$

Note that, for any $f, g, h \in L^p(\mu)$ and $1 \leq p \leq \infty$,

- (i) $d_p(f, g) = d_p(g, f) \geq 0$ (*nonnegativity and symmetry*), and
- (ii) $d_p(f, h) \leq d_p(f, g) + d_p(g, h)$ (*triangle inequality*),

which follows by Minkowski's inequality (Theorem 3.1.14) for $1 \leq p < \infty$, and by Proposition 3.2.1 for $p = \infty$. However, $d_p(f, g) = 0$ implies only that $f = g$ a.e. (μ) . Thus, $d_p(\cdot, \cdot)$ of (2.3) satisfies conditions (i) and (ii) of being a metric and it satisfies condition (iii) as well, provided any two functions f and g that agree a.e. (μ) are regarded as the same element of $L^p(\mu)$. This leads to the following:

Definition 3.2.2: For f, g in $L^p(\mu)$, f is called *equivalent* to g and is written as $f \sim g$, if $f = g$ a.e. (μ) .

It is easy to verify that the relation $\sim$ of Definition 3.2.2 is an *equivalence relation*, i.e., it satisfies

- (i) $f \sim f$ for all f in $L^p(\mu)$ (*reflexive*)
- (ii) $f \sim g \Rightarrow g \sim f$ (*symmetry*)
- (iii) $f \sim g, g \sim h \Rightarrow f \sim h$ (*transitive*).

This equivalence relation $\sim$ divides $L^p(\mu)$ into disjoint equivalence classes such that in each class all elements are equivalent. The notion of distance between these classes may be defined as follows:

$$d_p([f], [g]) \equiv d_p(f, g)$$

where $[f]$ and $[g]$ denote, respectively, the equivalence classes of functions containing f and g . It can be verified that this is a metric on the set of equivalence classes. In what follows, the equivalence class $[f]$ is identified with the element f . With this identification, $(L^p(\mu), d_p(\cdot, \cdot))$ becomes a metric space for $1 \leq p \leq \infty$.

Remark 3.2.1: For $0 < p < 1$, if one defines

$$d_p(f, g) \equiv \int |f - g|^p d\mu, \quad (2.4)$$

then $(L^p(\mu), d_p)$ becomes a metric space (with the same identification as above of functions with their equivalence classes). The triangle inequality follows from (2.2).

Recall that a metric space $(\mathbb{S}, d)$ is called *complete* if every *Cauchy sequence* in $(\mathbb{S}, d)$ converges to an element in $\mathbb{S}$, i.e., if $\{x_n\}_{n \geq 1}$ is a sequence in $\mathbb{S}$ such that for every $\epsilon > 0$, there exists a N_ϵ such that $n, m \geq N_\epsilon \Rightarrow d(x_n, x_m) \leq \epsilon$, then there exists an element x in $(\mathbb{S}, d)$ such that $\lim_{n \rightarrow \infty} d(x_n, x) = 0$. The next step is to establish the *completeness* of $L^p(\mu)$.

Theorem 3.2.2: For $0 < p \leq \infty$, $(L^p(\mu), d_p(\cdot, \cdot))$ is complete, where d_p is as in (2.3).

Proof: Let $\{f_n\}_{n \geq 1}$ be a Cauchy sequence in $L^p(\mu)$ for $0 < p < \infty$. The main steps in the proof are as follows:

- (I) there exists a subsequence $\{n_k\}_{k \geq 1}$ such that $\{f_{n_k}\}_{k \geq 1}$ converges a.e. (μ) to a limit function f ;
- (II) $\lim_{k \rightarrow \infty} d_p(f_{n_k}, f) = 0$;
- (III) $\lim_{n \rightarrow \infty} d_p(f_n, f) = 0$.

STEP (I): Let $\{\epsilon_k\}_{k \geq 1}$ and $\{\delta_k\}_{k \geq 1}$ be sequences of positive numbers decreasing to zero. Since $\{f_n\}_{n \geq 1}$ is Cauchy, for each $k \geq 1$, there exists an integer n_k such that

$$\int |f_n - f_m|^p d\mu \leq \epsilon_k \quad \text{for all } n, m \geq n_k. \quad (2.5)$$

W.l.o.g., let $n_{k+1} > n_k$ for each $k \geq 1$. Then, by Markov's inequality (Theorem 3.1.1),

$$\mu(\{|f_{n_{k+1}} - f_{n_k}| \geq \delta_k\}) \leq \delta_k^{-p} \int |f_{n_{k+1}} - f_{n_k}|^p d\mu \leq \delta_k^{-p} \epsilon_k. \quad (2.6)$$

Let $A_k = \{|f_{n_{k+1}} - f_{n_k}| \geq \delta_k\}$, $k = 1, 2, \dots$ and $A = \limsup_{k \rightarrow \infty} A_k \equiv \bigcap_{j=1}^{\infty} \bigcup_{k \geq j} A_k$. If $\{\epsilon_k\}_{k \geq 1}$ and $\{\delta_k\}_{k \geq 1}$ satisfy

$$\sum_{k=1}^{\infty} \delta_k^{-p} \epsilon_k < \infty, \quad (2.7)$$

then by (2.6), $\sum_{k=1}^{\infty} \mu(A_k) < \infty$ and hence, as in the proof of Theorem 2.5.2, $\mu(A) = 0$.

Note that for ω in A^c , $|f_{n_{k+1}}(\omega) - f_{n_k}(\omega)| < \delta_k$ for all k large. Thus, if $\sum_{k=1}^{\infty} \delta_k < \infty$, then for ω in A^c , $\{f_{n_k}(\omega)\}_{k \geq 1}$ is a Cauchy sequence in $\mathbb{R}$ and hence, it converges to some $f(\omega)$ in $\mathbb{R}$. Setting $f(\omega) = 0$ for $\omega \in A$, one gets

$$\lim_{k \rightarrow \infty} f_{n_k} = f \quad \text{a.e. } (\mu).$$

A choice of $\{\epsilon_k\}_{k \geq 1}$ and $\{\delta_k\}_{k \geq 1}$ such that $\sum_{k=1}^{\infty} \delta_k < \infty$ and (2.7) holds is given by $\epsilon_k = 2^{-(p+1)k}$ and $\delta_k = 2^{-k}$. This completes Step (I).

STEP (II): By Fatou's lemma, part (I), and (2.5), for any $k \geq 1$ fixed,

$$\epsilon_k \geq \liminf_{j \rightarrow \infty} \int |f_{n_k} - f_{n_{k+j}}|^p d\mu \geq \int |f_{n_k} - f|^p d\mu.$$

Since $f_{n_k} \in L^p(\mu)$, this shows that $f \in L^p(\mu)$. Now, on letting $k \rightarrow \infty$, (II) follows.

STEP (III): By triangle inequality, for any $k \geq 1$ fixed,

$$d_p(f_n, f) \leq d_p(f_n, f_{n_k}) + d_p(f_{n_k}, f).$$

By (2.5) and (II), for $n \geq n_k$, the right side above is $\leq 2\tilde{\epsilon}_k$, where $\tilde{\epsilon}_k = \epsilon_k$ if $0 < p < 1$ and $\tilde{\epsilon}_k = \epsilon_k^{1/p}$ if $1 \leq p < \infty$. Now letting $k \rightarrow \infty$, (III) follows.

The proof of Theorem 3.2.2 is complete for $0 < p < \infty$. The case $p = \infty$ is left as an exercise (Problem 3.14). $\square$

3.2.2 Dual spaces Set of all linear functionals on L^p functional: function on L^p space w/ number as output

Let $1 \leq p < \infty$. Let $g \in L^q(\mu)$, where $q = \frac{p}{p-1}$ if $1 < p < \infty$ and $q = \infty$ if $p = 1$. Let

$$T_g(f) = \int fg d\mu, \quad f \in L^p(\mu). \quad (2.8)$$

By Hölder's inequality, $\int |fg| d\mu < \infty$ and so $T_g(\cdot)$ is well defined. Clearly T_g is linear, i.e.,

$$T_g(\alpha_1 f_1 + \alpha_2 f_2) = \alpha_1 T_g(f_1) + \alpha_2 T_g(f_2) \quad (2.9)$$

for all $\alpha_1, \alpha_2 \in \mathbb{R}$ and $f_1, f_2 \in L^p(\mu)$.

Definition 3.2.3:

- (a) A function $T : L^p(\mu) \rightarrow \mathbb{R}$ that satisfies (2.9) is called a *linear functional*.
- (b) A linear functional T on $L^p(\mu)$ is called *bounded* if there is a constant $c \in (0, \infty)$ such that

$$|T(f)| \leq c \|f\|_p \quad \text{for all } f \in L^p(\mu).$$

- (c) The norm of a bounded linear functional T on $L^p(\mu)$ is defined as

$$\|T\| = \sup \{ |Tf| : f \in L^p(\mu), \|f\|_p = 1 \}.$$

By Hölder's inequality (cf. Theorem 3.1.11 and Remark 3.1.4),

$$|T_g(f)| \leq \|g\|_q \|f\|_p \quad \text{for all } f \in L^p(\mu),$$

and hence, T_g is a bounded linear functional on $L^p(\mu)$. This implies that if $d_p(f_n, f) \rightarrow 0$, then

$$|T_g(f_n) - T_g(f)| \leq \|g\|_q d_p(f_n, f) \rightarrow 0,$$

i.e., T_g is *continuous* on the metric space $(L^p(\mu), d_p)$.

Definition 3.2.4: The set of all continuous linear functionals on $L^p(\mu)$ is called the *dual space* of $L^p(\mu)$ and is denoted by $(L^p(\mu))^*$.

In the next section, it will be shown that continuity of a linear functional on $L^p(\mu)$ implies boundedness. A natural question is whether every continuous linear functional T on $L^p(\mu)$ coincides with T_g for some g in $L^q(\mu)$. The answer is “yes” for $1 \leq p < \infty$, as shown by the following result.

Theorem 3.2.3: (*Riesz representation theorem*). Let $1 \leq p < \infty$. Let $T : L^p(\mu) \rightarrow \mathbb{R}$ be linear and continuous. Then, there exists a g in $L^q(\mu)$ such that $T = T_g$, i.e.,

$$T(f) = T_g(f) \equiv \int f g d\mu \quad \text{for all } f \in L^p(\mu), \quad (2.10)$$

where $q = \frac{p}{p-1}$ for $1 < p < \infty$ and $q = \infty$ if $p = 1$.

Remark 3.2.2: Such a representation is not valid for $p = \infty$. That is, there exists continuous linear functionals T on $L^\infty(\mu)$ for which there is no $g \in L^1(\mu)$ such that $T(f) = \int f g d\mu$ for all $f \in L^\infty(\mu)$, provided μ is not concentrated on a finite set $\{\omega_1, \omega_2, \dots, \omega_k\} \subset \Omega$.

For a proof of Theorem 3.2.3 and the above remark, see Royden (1988) or Rudin (1987).

Next consider the mapping from $(L^p(\mu))^*$ and $L^q(\mu)$ defined by

$$\phi(T_g) = g,$$

where T_g is as defined in (2.10). Then, ϕ is linear, i.e.,

$$\phi(\alpha_1 T_1 + \alpha_2 T_2) = \alpha_1 \phi(T_1) + \alpha_2 \phi(T_2)$$

for all $\alpha_1, \alpha_2 \in \mathbb{R}$ and $T_1, T_2 \in (L^p(\mu))^*$. Further,

$$\|\phi(T)\|_q = \|T\| \quad \text{for all } T \in (L^p(\mu))^*.$$

Thus, ϕ preserves the vector space structure of $(L^p(\mu))^*$ and the norm. For this reason, it is called an *isometry* between $(L^p(\mu))^*$ and $L^q(\mu)$.

3.3 Banach and Hilbert spaces

3.3.1 Banach spaces

If $(\Omega, \mathcal{F}, \mu)$ is a measure space it was seen in the previous section that the space $L^p(\Omega, \mathcal{F}, \mu)$ of equivalence classes of functions f with $\int_\Omega |f|^p d\mu < \infty$ is a vector space over $\mathbb{R}$ for all $1 \leq p < \infty$ and for $p \geq 1$, $\|\cdot\|_p \equiv (\int |f|^p d\mu)^{1/p}$ satisfies

very optional

- (i) $\|f + g\| \leq \|f\| + \|g\|$,
- (ii) $\|\alpha f\| = |\alpha| \|f\|$ for every $\alpha \in \mathbb{R}$,
- (iii) $\|f\| = 0$ iff $f = 0$ a.e. (μ) .

The Euclidean spaces $\mathbb{R}^k$ for any $k \in \mathbb{N}$ is also a vector space. Note that for $p \geq 1$, setting $\|x\|_p \equiv (\sum_{i=1}^k |x_i|^p)^{1/p}$ if $x = (x_1, x_2, \dots, x_k)$, $(\mathbb{R}^k, \|x\|_p)$ may be identified with a special case of $L^p(\Omega, \mathcal{F}, \mu)$, where $\Omega \equiv \{1, 2, \dots, k\}$, $\mathcal{F} = \mathcal{P}(\Omega)$ and μ is the counting measure.

Generalizing the above examples leads to the notion of a *normed vector space* (also called *normed linear space*). Recall that a *vector space* V over $\mathbb{R}$ is a nonempty set with a binary operation $+$, a function from $V \times V$ to V (called *addition*), and scalar multiplication by the real numbers, i.e., a function from $\mathbb{R} \times V \rightarrow V$, $((\alpha, v) \rightarrow \alpha v)$ satisfying

- (i) $v_1, v_2 \in V \Rightarrow v_1 + v_2 = v_2 + v_1 \in V$.
- (ii) $v_1, v_2, v_3 \in V \Rightarrow (v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$.
- (iii) There exists an element θ , called the *zero vector*, in V such that $v + \theta = v$ for all v in V .
- (iv) $\alpha \in \mathbb{R}, v \in V \Rightarrow \alpha v \in V$.
- (v) $\alpha \in \mathbb{R}, v_1, v_2 \in V \Rightarrow \alpha(v_1 + v_2) = \alpha v_1 + \alpha v_2$.
- (vi) $\alpha_1, \alpha_2 \in \mathbb{R}, v \in V \Rightarrow (\alpha_1 + \alpha_2)v = \alpha_1 v + \alpha_2 v$ and $\alpha_1(\alpha_2 v) = (\alpha_1 \alpha_2)v$.
- (vii) $v \in V \Rightarrow 0v = \theta$ and $1v = v$.

Note that from conditions (vi) and (vii) above, it follows that for any v in V , $v + (-1)v = 0 \cdot v = \theta$. Thus for any v in V , $(-1)v$ is the additive inverse and is denoted by $-v$. Conditions (i), (ii), and (iii) are called respectively *commutativity*, *associativity*, and the *existence of an additive identity*. Thus V under the operation $+$ is an Abelian (i.e., commutative) group.

Definition 3.3.1: A function f from V to $\mathbb{R}_+$ denoted by $f(v) \equiv \|v\|$ is called a *norm* if

- (a) $v_1, v_2 \in V \Rightarrow \|v_1 + v_2\| \leq \|v_1\| + \|v_2\|$ (*triangle inequality*)
- (b) $\alpha \in \mathbb{R}, v \in V \Rightarrow \|\alpha v\| = |\alpha| \|v\|$ (*scalar homogeneity*)
- (c) $\|v\| = 0$ iff $v = \theta$.

A vector space V with a norm $\|\cdot\|$ defined on it is called a *normed vector space* or *normed linear space* and is denoted as $(V, \|\cdot\|)$. Let $d(v_1, v_2) \equiv \|v_1 - v_2\|$, $v_1, v_2 \in V$. Then from the definition of $\|\cdot\|$, it follows that d is a metric on V , i.e., (V, d) is a metric space. Recall that a metric space (S, d)

is called *complete* if every Cauchy sequence $\{x_n\}_{n \geq 1}$ in $\mathbb{S}$ converges to an element x in $\mathbb{S}$.

Definition 3.3.2: A *Banach space* is a complete normed linear space $(V, \|\cdot\|)$.

It was shown by S. Banach of Poland that all $L^p(\Omega, \mathcal{B}, \mu)$ spaces are Banach spaces, provided $p \geq 1$ and in particular, all Euclidean spaces are Banach spaces. An example of a different kind is the space $C[0, 1]$ of all real valued continuous functions on $[0, 1]$ with the usual operation of pointwise addition and scalar multiplication, i.e., $(f + g)(x) = f(x) + g(x)$ and $(\alpha f)(x) = \alpha \cdot f(x)$ for all $\alpha \in \mathbb{R}$, $0 \leq x \leq 1$, $f, g \in C[0, 1]$ where the norm (called the *supnorm*) is defined by $\|f\| = \sup\{|f(x)| : 0 \leq x \leq 1\}$. The verification of the fact that $C[0, 1]$ with the supnorm is a Banach space is left as an exercise (Problem 3.22). The space $\mathcal{P}$ of all polynomials on $[0, 1]$ is also a normed linear space under the above norm but $(\mathcal{P}, \|\cdot\|)$ is *not complete* (Problem 3.23). However for each $n \in \mathbb{N}$, the space $\mathcal{P}_n$ of all polynomials on $[0, 1]$ of degree $\leq n$ is a Banach space under the supnorm (Problem 3.26).

Definition 3.3.3: Let V be a vector space. A subset $W \subset V$ is called a *subspace* of V if $v_1, v_2 \in W, \alpha_1, \alpha_2 \in \mathbb{R} \Rightarrow \alpha_1 v_1 + \alpha_2 v_2 \in W$. If $(V, \|\cdot\|)$ is a normed vector space and W is a subspace of V , then $(W, \|\cdot\|)$ is also a normed vector space. If W is closed in $(V, \|\cdot\|)$, then W is called a *closed subspace* of V .

Remark 3.3.1: If $(V, \|\cdot\|)$ is a Banach space and W is a closed subspace of V , then $(W, \|\cdot\|)$ is also a Banach space.

3.3.2 Linear transformations

Let $(V_i, \|\cdot\|_i)$, $i = 1, 2$ be two normed linear spaces over $\mathbb{R}$.

Definition 3.3.4: A function T from V_1 to V_2 is called a *linear transformation* or *linear operator* if $\alpha_1, \alpha_2 \in \mathbb{R}, x, y \in V_1 \Rightarrow T(\alpha_1 x + \alpha_2 y) = \alpha_1 T(x) + \alpha_2 T(y)$.

Definition 3.3.5: A linear operator T from $(V_1, \|\cdot\|_1)$ to $(V_2, \|\cdot\|_2)$ is called *bounded* if $\|T\| \equiv \sup\{\|Tx\|_2 : \|x\|_1 < 1\} < \infty$, i.e., the image of the *unit ball* in $(V_1, \|\cdot\|_1)$ is contained in a ball of finite radius centered at the zero in V_2 .

Here is a summary of some important results on this topic. By linearity of T , $T(\frac{x}{\|x\|}) = \frac{1}{\|x\|}T(x)$ for any $x \neq 0$. It follows that T is *bounded* iff there exists $k < \infty$ such that for any $x \in V_1$, $\|Tx\|_2 \leq k\|x\|_1$. Clearly, then k can be taken to be $\|T\|$. Also by linearity, if T is bounded, then $\|Tx_1 - Tx_2\| = \|T(x_1 - x_2)\| \leq \|T\| \|x_1 - x_2\|$ and so the map T is *continuous*.

(indeed, uniformly so). It turns out that if a linear operator T is continuous at some x_0 in V_1 , then T is continuous on all of V_1 and is bounded (Problem 3.28 (a)).

Now let $B(V_1, V_2)$ be the space of all bounded linear operators from $(V_1, \|\cdot\|_1)$ to $(V_2, \|\cdot\|_2)$. For $T_1, T_2 \in B(V_1, V_2)$, α_1, α_2 in $\mathbb{R}$, let $(\alpha_1 T_1 + \alpha_2 T_2)$ be defined by $(\alpha_1 T_1 + \alpha_2 T_2)(x) \equiv \alpha_1 T_1(x) + \alpha_2 T_2(x)$ for all x in V_1 . Then it can be verified that $(\alpha_1 T_1 + \alpha_2 T_2)$ also belongs to $B(V_1, V_2)$ and

$$\|T\| \equiv \sup\{\|Tx\|_2 : \|x\|_1 \leq 1\} \quad (3.1)$$

is a norm on $B(V_1, V_2)$. Thus $(B(V_1, V_2), \|\cdot\|)$ is also a normed linear space.

If $(V_2, \|\cdot\|_2)$ is complete, then it can be shown that $(B(V_1, V_2), \|\cdot\|)$ is also a Banach space (Problem 3.28 (b)). In particular, if $(V_2, \|\cdot\|_2)$ is the real line, the space $(B(V_1, \mathbb{R}), \|\cdot\|)$ is a Banach space.

3.3.3 Dual spaces

Definition 3.3.6: The space of all bounded linear functions from $(V_1, \|\cdot\|)$ to $\mathbb{R}$ (also called *bounded linear functionals*), denoted by V_1^* , is called the *dual space* of V_1 .

Thus, for any normed linear space $(V_1, \|\cdot\|_1)$ (that need not be complete), the dual space $(V_1^*, \|\cdot\|)$ is always a Banach space, where $\|T\| \equiv \sup\{|Tx| : \|x\|_1 < 1\}$ for $T \in V_1^*$. If $(V_1, \|\cdot\|_1) = L^p(\Omega, \mathcal{F}, \mu)$ for some measure space $(\Omega, \mathcal{F}, \mu)$ and $1 \leq p < \infty$, by the *Riesz representation theorem* (see Theorem 3.2.3), the dual space may be identified with $L^q(\Omega, \mathcal{F}, \mu)$ where q is the conjugate of p , i.e., $\frac{1}{p} + \frac{1}{q} = 1$. However, as pointed out earlier in Section 3.2, this is not true for $p = \infty$. That is, the dual of $L^\infty(\Omega, \mathcal{F}, \mu)$ is not $L^1(\Omega, \mathcal{F}, \mu)$ unless $(\Omega, \mathcal{F}, \mu)$ is a measure space where Ω is a finite set $\{w_1, w_2, \dots, w_k\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$. An example for the $p = \infty$ case can be constructed for the space ℓ_∞ of all bounded sequences of real numbers (cf. Royden (1988)).

The representation of the dual space of the Banach space $C[0, 1]$ with supnorm is in terms of finite signed measures (cf. Section 4.2).

Theorem 3.3.1: (*Riesz*). Let $T : C[0, 1] \rightarrow \mathbb{R}$ be linear and bounded. Then there exists two finite measures μ_1 and μ_2 on $[0, 1]$ such that for any $f \in C[0, 1]$

$$T(f) = \int f d\mu_1 - \int f d\mu_2.$$

For a proof see Royden (1988) or Rudin (1987) (see also Problem 3.27).

3.3.4 Hilbert space

A vector space V over $\mathbb{R}$ is called a *real innerproduct space* if there exists a function $f : V \times V \rightarrow \mathbb{R}$, denoted by $f(x, y) \equiv \langle x, y \rangle$ (and called the *innerproduct*) that satisfies

- (i) $\langle x, y \rangle = \langle y, x \rangle$ for all $x, y \in V$,
- (ii) (*linearity*) $\langle \alpha_1 x_1 + \alpha_2 x_2, y \rangle = \alpha_1 \langle x_1, y \rangle + \alpha_2 \langle x_2, y \rangle$ for all $\alpha_1, \alpha_2 \in \mathbb{R}$, $x_1, x_2, y \in V$,
- (iii) $\langle x, x \rangle \geq 0$ for all $x \in V$ and $\langle x, x \rangle = 0$ iff $x = \theta$, the zero vector of V .

Using the fact that the quadratic function $\varphi(t) = \langle x + ty, x + ty \rangle = \langle x, x \rangle + 2t\langle x, y \rangle + t^2\langle y, y \rangle$ is nonnegative for all $t \in \mathbb{R}$, one gets the *Cauchy-Schwarz inequality*

$$|\langle x, y \rangle| \leq \sqrt{\langle x, x \rangle \langle y, y \rangle} \quad \text{for all } x, y \in V.$$

Now setting $\|x\| = \sqrt{\langle x, x \rangle}$ and using the Cauchy-Schwarz inequality, one verifies that $\|x\|$ is a norm on V and thus $(V, \|\cdot\|)$ is a normed linear space. Further, the function $\langle x, y \rangle$ from $V \times V$ to $\mathbb{R}$ is continuous (Problem 3.29) under the norm $\|(x_1, x_2)\| = \|x_1\| + \|x_2\|$, $(x_1, x_2) \in V \times V$.

Definition 3.3.7: Let $(V, \langle \cdot, \cdot \rangle)$ be a real innerproduct space. It is called a *Hilbert space* if $(V, \|\cdot\|)$ is a Banach space, i.e., if it is complete.

It was seen in Section 3.2 that for any measure space $(\Omega, \mathcal{F}, \mu)$, the space $L^2(\Omega, \mathcal{F}, \mu)$ of all equivalence classes of functions $f : \Omega \rightarrow \mathbb{R}$ satisfying $\int |f|^2 d\mu < \infty$ is a complete innerproduct space with the innerproduct $\langle f, g \rangle = \int f g d\mu$ and hence a Hilbert space. It turns out that every Hilbert space H is an $L^2(\Omega, \mathcal{F}, \mu)$ for some $(\Omega, \mathcal{F}, \mu)$. (The axiom of choice or its equivalent, the Hausdorff's maximality principle, is required for a proof of this. See Rudin (1987).) This is in contrast to the Banach space case where every $L^p(\Omega, \mathcal{F}, \mu)$ with $p \geq 1$ is a Banach space but not conversely, i.e., every Banach space need not be an $L^p(\Omega, \mathcal{F}, \mu)$.

Next for each x in a Hilbert space H , let $T_x : H \rightarrow \mathbb{R}$ be defined by $T_x(y) = \langle x, y \rangle$. By the defining properties of $\langle x, y \rangle$ and the Cauchy-Schwarz inequality, it is easy to verify that T_x is a bounded linear function on H , i.e.,

$$T_x(\alpha_1 y_1 + \alpha_2 y_2) = \alpha_1 T_x(y_1) + \alpha_2 T_x(y_2) \quad \text{for all } \alpha_1, \alpha_2 \in \mathbb{R}, y_1, y_2 \in H \quad (3.2)$$

and

$$|T_x(y)| \leq \|x\| \|y\| \quad \text{for all } y \in H. \quad (3.3)$$

Thus $T_x \in H^*$, the dual space. It is an important result (see Theorem 3.3.3 below) that every $T \in H^*$ is equal to T_x for some x in H and $\|T\| = \|x\|$. Thus H^* can be identified with H .

Definition 3.3.8: Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Two vectors x, y in V are said to be *orthogonal* and written as $x \perp y$ if $\langle x, y \rangle = 0$.

A collection $B \subset V$ is called *orthogonal* if $x, y \in B$, $x \neq y \Rightarrow \langle x, y \rangle = 0$. The collection B is called *orthonormal* if it is orthogonal and in addition for all x in B , $\|x\| = 1$. Note that if $x \perp y$, then $\|x - y\|^2 = \langle x - y, x - y \rangle = \langle x, x \rangle + \langle y, y \rangle = \|x\|^2 + \|y\|^2$ and so if B is an orthonormal set, then $x, y \in B \Rightarrow$ either $x = y$ or $\|x - y\| = \sqrt{2}$. Thus, if V is separable under the metric $d(x, y) = \|x - y\|$ (i.e., there exists a countable set $D \subset V$ such that for every x in V and $\epsilon > 0$, there exists a $d \in D$ such that $\|x - d\| < \epsilon$) and if $B \subset V$ is an orthonormal system, then the open ball S_b of radius $\frac{1}{2\sqrt{2}}$ around each $b \in B$ satisfies $\{S_b \cap D : b \in B\}$ are disjoint and nonempty. Thus B is countable.

Now let $(V, \langle \cdot, \cdot \rangle)$ be a separable innerproduct space and $B \subset V$ be an orthonormal system.

Definition 3.3.9: The *Fourier coefficients* of a vector x in V with respect on *orthonormal set* B is the set $\{\langle x, b \rangle : b \in B\}$.

Since V is separable, B is countable. Let $B = \{b_i : i \in \mathbb{N}\}$. For a given $x \in V$, let $c_i = \langle x, b_i \rangle$, $i \geq 1$. Let $x_n \equiv \sum_{i=1}^n c_i b_i$, $n \in \mathbb{N}$. The sequence $\{x_n\}_{n \geq 1}$ is called the *partial sum sequence of the Fourier expansion of the vector x w.r.t. the orthonormal set B* .

A natural question is: *when does $\{x_n\}_{n \geq 1}$ converge to x ?* By the linearity property in the definition of the innerproduct $\langle \cdot, \cdot \rangle$, it follows that

$$0 \leq \|x - x_n\|^2 = \langle x - x_n, x - x_n \rangle = \langle x, x \rangle - 2\langle x, x_n \rangle + \langle x_n, x_n \rangle$$

and

$$\langle x, x_n \rangle = \sum_{i=1}^n c_i \langle x, b_i \rangle = \sum_{i=1}^n c_i^2.$$

Since $\{b_i\}_{i \geq 1}$ are orthonormal,

$$\|x_n\|^2 = \langle x_n, x_n \rangle = \sum_{i=1}^n c_i^2 = \langle x, x_n \rangle.$$

Thus,

$$0 \leq \|x - x_n\|^2 = \|x\|^2 - \|x_n\|^2 = \|x\|^2 - \sum_{i=1}^n c_i^2,$$

leading to

Proposition 3.3.2: (*Bessel's inequality*). Let $\{b_i\}_{i \geq 1}$ be orthonormal in an innerproduct space $(V, \langle \cdot, \cdot \rangle)$. Then, for any x in V ,

$$\sum_{i=1}^{\infty} \langle x, b_i \rangle^2 \leq \|x\|^2. \quad (3.4)$$

Now let $(V, \langle \cdot, \cdot \rangle)$ be a *Hilbert space*. Since for $m > n$,

$$\|x_n - x_m\|^2 = \sum_{i=n+1}^m \langle x, b_i \rangle^2,$$

it follows from Bessel's inequality that $\{x_n\}_{n \geq 1}$ is a Cauchy sequence, and since V is complete, there is a y in V such that $x_n \rightarrow y$. This implies (by the continuity of $\langle x, y \rangle$) that $\langle x, b_i \rangle = \lim_{n \rightarrow \infty} \langle x_n, b_i \rangle = \langle y, b_i \rangle \Rightarrow \langle x - y, b_i \rangle = 0$ for all $i \geq 1$. Thus, it follows that $\langle x, b_i \rangle = \langle y, b_i \rangle$ for all $i \geq 1$. The last relation implies $y = x$ iff the set $\{b_i\}_{i \geq 1}$ satisfies the property that

$$\langle z, b_i \rangle = 0 \quad \text{for all } i \geq 1 \Rightarrow \|z\| = 0. \quad (3.5)$$

This property is called the *completeness of B* . Thus $B \equiv \{b_i\}_{i \geq 1}$ is a *complete orthonormal set* for a Hilbert space $H \equiv (V, \langle \cdot, \cdot \rangle)$, iff for every vector x ,

$$\sum_{i=1}^{\infty} c_i^2 = \|x\|^2, \quad (3.6)$$

where $c_i = \langle x, b_i \rangle$, $i \geq 1$, which in turn holds iff

$$\left\| x - \sum_{i=1}^n c_i b_i \right\| \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (3.7)$$

Conversely, if $\{c_i\}_{i \geq 1}$ is a sequence of real numbers such that $\sum_{i=1}^{\infty} c_i^2 < \infty$, then the sequence $\{x_n \equiv \sum_{i=1}^n c_i b_i\}_{n \geq 1}$ is Cauchy and hence converges to an x in V . Thus the Hilbert space H can be identified with the space ℓ_2 of all *square summable* sequences $\{\{c_i\}_{i \geq 1} : \sum_{i=1}^{\infty} c_i^2 < \infty\}$, in the sense that the map $\varphi : x \rightarrow \{c_i\}_{i \geq 1}$, where $c_i = \langle x, b_i \rangle$, $i \geq 1$, preserves the algebraic structure as well as the innerproduct, i.e., φ is a linear operator from H to ℓ_2 and $\langle \varphi(x), \varphi(y) \rangle = \langle x, y \rangle$ for all $x, y \in H$. Such a φ is called an *isometric isomorphism* between H to ℓ_2 . Note also that ℓ_2 is simply $L_2(\Omega, \mathcal{F}, \mu)$ where $\Omega \equiv \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, and μ , the counting measure. It can be shown (using the axiom of choice) that every separable Hilbert space does possess a complete orthonormal system, i.e., an *orthonormal basis*.

Next some examples are given. Here, unless otherwise indicated, H denotes the Hilbert space and B denotes an orthonormal basis of H .

Example 3.3.1:

- (a) $H \equiv \ell_2 = \{(x_1, x_2, \dots) : x_i \in \mathbb{R}, \sum_{i=1}^{\infty} x_i^2 < \infty\}$.
 $B \equiv \{e_i : i \geq 1\}$ where $e_i \equiv (0, 0, \dots, 1, 0, \dots)$ with 1 in the i th position and 0 elsewhere.
- (b) $H \equiv L^2([0, 2\pi], \mathcal{B}([0, 2\pi]), \mu)$ where $\mu(A) = \frac{1}{2\pi} m(A)$, $m(\cdot)$ being Lebesgue measure.
 $B \equiv \{\cos nx : n = 0, 1, 2, \dots\} \cup \{\sin nx : n = 1, 2, \dots\}$. (For a proof, see Chapter 5.)

(c) Let $H \equiv L^2(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$ where μ is a finite measure such that $\int |x|^k d\mu < \infty$ for all $k = 1, 2, \dots$.

Let $B_1 \equiv \{1, x, x^2, \dots\}$ and B be the orthonormal set generated by applying the Gram-Schmidt procedure to B_1 (see Problem 3.31). It can be shown that B is a basis for H (Problem 3.39). When μ is the standard normal distribution, the elements of B are called *Hermite* polynomials.

For one more example, i.e., Haar functions, see Problem 3.40.

Theorem 3.3.3: (*Riesz representation*). *Let H be a separable Hilbert space. Then every bounded linear functional T on $H \rightarrow \mathbb{R}$ can be represented as $T \equiv T_{x_0}$ for some $x_0 \in H$, where $T_{x_0}(y) \equiv \langle y, x_0 \rangle$.*

Proof: Let $B = \{b_i\}_{i \geq 1}$ be an orthonormal basis for H . Let $c_i \equiv T(b_i)$, $i \geq 1$. Then, for $n \geq 1$,

$$\begin{aligned} \sum_{i=1}^n c_i^2 &= \sum_{i=1}^n c_i T(b_i) \\ &= T\left(\sum_{i=1}^n c_i b_i\right) \quad (\text{by the linearity of } T) \\ \Rightarrow \left|\sum_{i=1}^n c_i^2\right| &\leq \|T\| \left\|\sum_{i=1}^n c_i b_i\right\| = \|T\| \left(\sum_{i=1}^n c_i^2\right)^{1/2} \\ \Rightarrow \sum_{i=1}^n c_i^2 &\leq \|T\|^2 \\ \Rightarrow \sum_{i=1}^{\infty} c_i^2 &< \infty. \end{aligned}$$

Thus $\{x_n \equiv \sum_{i=1}^n c_i b_i\}_{n \geq 1}$ is Cauchy in H and hence converges to an x_0 in H . By the continuity of T , for any y , $Ty = \lim_{n \rightarrow \infty} Ty_n$, where $y_n \equiv \sum_{i=1}^n \langle y, b_i \rangle b_i$, $n \geq 1$. But

$$\begin{aligned} Ty_n &= \sum_{i=1}^n \langle y, b_i \rangle c_i = \left\langle y, \sum_{i=1}^n b_i c_i \right\rangle \\ &= \langle y, x_n \rangle, \quad \text{by the linearity of } T \end{aligned}$$

Again by continuity of $\langle y, x \rangle$, it follows that $Ty = \langle y, x_0 \rangle$. $\square$

A sufficient condition for an $L^2(\Omega, \mathcal{F}, \mu)$ to be separable is that there exists an at most countable family $\mathcal{A} \equiv \{A_j\}_{j \geq 1}$ of sets in $\mathcal{F}$ such that $\mathcal{F} = \sigma(\mathcal{A})$ and $\mu(A_j) > 0$ for each j . This holds for any σ -finite measure μ on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ (Problem 3.38).

Remark 3.3.2: Assuming the axiom of choice, the Riesz representation theorem remains valid for any Hilbert space, separable or not (Problem 3.43).

3.4 Problems

3.1 Prove Proposition 3.1.7.

(**Hint:** Use (1.4) repeatedly.)

3.2 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space with $\mu(\Omega) \leq 1$ and $f : \Omega \rightarrow (a, b) \subset \mathbb{R}$ be in $L^1(\Omega, \mathcal{F}, \mu)$. Let $\phi : (a, b) \rightarrow \mathbb{R}$ be convex. Show that if $c \equiv \int f d\mu \in (a, b)$ and $\phi(f) \in L^1(\Omega, \mathcal{F}, \mu)$ and $c\phi'_+(c) \geq 0$, then

$$\mu(\Omega) \phi\left(\int f d\mu\right) \leq \int \phi(f) d\mu.$$

3.3 Prove Proposition 3.1.10.

(**Hint:** Apply Jensen's inequality with $\Omega \equiv \{1, 2, \dots, k\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, $P(\{i\}) = p_i$, $f(i) = a_i$, $i = 1, 2, \dots, k$, and $\phi(x) = e^x$ to get (i). Deduce (ii) from (i) and Remark 3.1.3. For (iii), consider $\phi(x) = |x|^p$.)

3.4 Give an example of a convex function ϕ on $(0, 1)$ with a finite number of points where it is not differentiable. Can this be extended to the countable case? Uncountable case?

(**Hint:** Note that $\phi'_+(\cdot)$ and $\phi'_-(\cdot)$ are both monotone and hence have at most a countable number of discontinuity points.)

3.5 Let $\phi : (a, b) \rightarrow \mathbb{R}$ be convex.

(a) Using the definition and induction, show that

$$\phi\left(\sum_{i=1}^n p_i x_i\right) \leq \sum_{i=1}^n p_i \phi(x_i)$$

for any $n \geq 2$, $x_1, x_2, \dots, x_n$ in (a, b) and $\{p_1, p_2, \dots, p_n\}$, a probability distribution.

(b) Use (a) to prove Jensen's inequality for any bounded ϕ .

3.6 Show that a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex iff

$$\phi\left(\int_{[0,1]} f dm\right) \leq \int_{[0,1]} \phi(f) dm$$

for every bounded Borel measurable function $f : [0, 1] \rightarrow \mathbb{R}$, where $m(\cdot)$ is the Lebesgue measure.

3.7 Let ϕ be convex on (a, b) and $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be convex and nondecreasing. Show that $\psi \circ \phi$ is convex on (a, b) .

3.8 Let X be a nonnegative random variable on some probability space.

(a) Show that $(EX)(E\frac{1}{X}) \geq 1$. What does this say about the correlation between X and $\frac{1}{X}$?

(b) Let $f, g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be Borel measurable and such that $f(x)g(x) \geq 1$ for all x in $\mathbb{R}_+$. Show that $Ef(X)Eg(X) \geq 1$.

3.9 Prove Corollary 3.1.13 using Hölder's inequality applied to an appropriate measure space.

3.10 Extend Hölder's inequality as follows. Let $1 < p_i < \infty$, and $f_i \in L^{p_i}(\Omega, \mathcal{F}, \mu)$, $i = 1, 2, \dots, k$. Suppose $\sum_{i=1}^k \frac{1}{p_i} = 1$. Show that $\int (\prod_{i=1}^k f_i) d\mu \leq \prod_{i=1}^k \|f_i\|_{p_i}$.

(**Hint:** Use Proposition 3.1.10 (ii).)

3.11 Verify Minkowski's inequality for $p = 1$ and $p = \infty$.

3.12 (a) Find $(\Omega, \mathcal{F}, \mu)$, $0 < p < 1$, $f, g \in L^p(\Omega, \mathcal{F}, \mu)$ such that

$$\left(\int |f + g|^p d\mu \right)^{1/p} > \left(\int |f|^p d\mu \right)^{1/p} + \left(\int |g|^p d\mu \right)^{1/p}.$$

(b) Prove (1.18) for $0 < p < 1$ with $\|f\|_p = \left(\int |f|^p d\mu \right)^{1/p}$.

3.13 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Let $\{A_k\}_{k \geq 1} \subset \mathcal{F}$ and $\sum_{k=1}^{\infty} \mu(A_k) < \infty$. Show that $\mu\left(\overline{\lim_{k \rightarrow \infty} A_k}\right) = 0$, where $\overline{\lim_{k \rightarrow \infty} A_k} = \bigcap_{n=1}^{\infty} \bigcup_{j \geq n} A_j = \{\omega : \omega \in A_j \text{ for infinitely many } j \geq 1\}$.

3.14 Establish Theorem 3.2.2 for $p = \infty$.

(**Hint:** For each $k \geq 1$, choose $n_k \uparrow$ such that $\|f_{n_{k+1}} - f_{n_k}\|_{\infty} < 2^{-k}$. Show that there exists a set A with $\mu(A^c) = 0$ and for ω in A , $|f_{n_{k+1}}(\omega) - f_{n_k}(\omega)| < 2^{-k}$ for all $k \geq 1$ and now proceed as in the proof for the case $0 < p < \infty$.)

3.15 Let $f, g \in L^p(\Omega, \mathcal{F}, \mu)$, $0 < p < 1$. Show that $d(f, g) = \int |f - g|^p d\mu$ is a metric and $(L^p(\Omega, \mathcal{F}, \mu), d)$ is complete.

3.16 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $f : \Omega \rightarrow \mathbb{R}$ be $\mathcal{F}$ -measurable. Let $A_f = \{p : 0 < p < \infty, \int |f|^p d\mu < \infty\}$.

(a) Show that $p_1, p_2 \in A_f$, $p_1 < p_2$ implies $[p_1, p_2] \subset A_f$.

(**Hint:** Use $\int_{|f| \geq 1} |f|^p d\mu \leq \int_{|f| \geq 1} |f|^{p_2} d\mu$ and $\int_{|f| \leq 1} |f|^p d\mu \leq \int_{|f| \leq 1} |f|^{p_1} d\mu$ for any $p_1 < p < p_2$.)

- (b) Let $\psi(p) = \log \int |f|^p d\mu$ for $p \in A_f$. By (a), it is known that A_f is connected, i.e., it is an interval. Show that ψ is convex in the open interior of A_f .

(**Hint:** Use Hölder's inequality.)

- (c) Give examples to show that A_f could be a closed interval, an open interval, and semi-open intervals.
- (d) If $0 < p_1 < p < p_2$, show that

$$\|f\|_p \leq \max\{\|f\|_{p_1}, \|f\|_{p_2}\}$$

(**Hint:** Use (b).)

- (e) Show that if $\int |f|^r d\mu < \infty$ for some $0 < r < \infty$, then $\|f\|_p \rightarrow \|f\|_\infty$ as $p \rightarrow \infty$.

(**Hint:** Show first that for any $K > 0$, $\mu(|f| > K) > 0 \Rightarrow \lim_{p \rightarrow \infty} \|f\|_p \geq K$. If $\|f\|_\infty < \infty$ and $\mu(\Omega) < \infty$, use the fact that $\|f\|_p \leq \|f\|_\infty (\mu(\Omega))^{1/p}$ and reduce the general case under the hypothesis that $\int |f|^p d\mu < \infty$ for some p to this case.)

3.17 Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mu)$. Recall that $Eh(X) = \int h(X) d\mu$ if $h(X) \in L^1(\Omega, \mathcal{F}, \mu)$.

- (a) Show that $(E|X|^{p_1}) \leq (E|X|^{p_2})^{\frac{p_1}{p_2}}$ for any $0 < p_1 < p_2 < \infty$.
- (b) Show that '=' holds in (a) iff $|X|$ is a constant a.e. (μ) .
- (c) Show that if $E|X| < \infty$, then $E|\log |X|| < \infty$ and $E|X|^r < \infty$ for all $0 < r < 1$, and $\frac{1}{r} \log(E|X|^r) \rightarrow E \log |X|$ as $r \rightarrow 0$.

3.18 Let X be a nonnegative random variable.

- (a) Show that $EX \log X \geq (EX)(E \log X)$.
- (b) Show that $\sqrt{1 + (EX)^2} \leq E(\sqrt{1 + X^2}) \leq 1 + EX$.

3.19 Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, and let μ be the counting measure. Denote $L^p(\Omega, \mathcal{F}, \mu)$ for this case by ℓ_p .

- (a) Show that ℓ_p is the set of all sequences $\{x_n\}_{n \geq 1}$ such that $\sum_{n=1}^{\infty} |x_n|^p < \infty$.
- (b) For the following sequences, find all $p > 0$ such that they belong to ℓ_p :
- (i) $x_n \equiv \frac{1}{n}$, $n \geq 1$.
- (ii) $x_n = \frac{1}{n(\log(n+1))^2}$, $n \geq 1$.

- 3.20 For $1 \leq p < \infty$, prove the Riesz representation theorem for ℓ_p . That is, show that if T is a bounded linear functional from $\ell_p \rightarrow \mathbb{R}$, then there exists a $y = \{y_i\}_{i \geq 1} \in \ell_q$ such that for any $x = \{x_i\}_{i \geq 1}$ in ℓ_p , $T(x) = \sum_{i=1}^{\infty} x_i y_i$.

(**Hint:** Let $y_i = T(e_i)$ where $e_i = \{e_i(j)\}_{j \geq 1}$, $e_i(j) = 1$ if $i = j$, 0 if $i \neq j$. Use the fact $|T(x)| \leq \|T\| \|x\|_p$ to show that for each $n \in \mathbb{N}$, $(\sum_{i=1}^n |y_i|^q) \leq \|T\|^q < \infty$.)

- 3.21 Let $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$, $\mu = \mu_F$ where F is a cdf on $\mathbb{R}$. If $f(x) \equiv x^2$, find $A_f = \{p : 0 < p < \infty, f \in L^p(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F)\}$ for the following cases:

(a) $F(x) = \Phi(x)$, the $N(0, 1)$ cdf, i.e., $\Phi(x) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-u^2/2} du$, $x \in \mathbb{R}$.

(b) $F(x) = \frac{1}{\pi} \int_{-\infty}^x \frac{1}{1+u^2} du$, $x \in \mathbb{R}$.

- 3.22 Show that $C[0, 1]$ with the supnorm (i.e., with $\|f\| = \sup\{|f(x)| : 0 \leq x \leq 1\}$) is a Banach space.

(**Hint:** To verify completeness, let $\{f_n\}_{n \geq 1}$ be a Cauchy sequence in $C[0, 1]$. Show that for each $0 \leq x \leq 1$, $\{f_n(x)\}_{n \geq 1}$ is a Cauchy sequence in $\mathbb{R}$. Let $f(x) = \lim_{n \rightarrow \infty} f_n(x)$. Now show that $\sup\{|f_n(x) - f(x)| : 0 \leq x \leq 1\} \leq \lim_{m \rightarrow \infty} \|f_n - f_m\|$. Conclude that f_n converges to f uniformly on $[0, 1]$ and that $f \in C[0, 1]$.)

- 3.23 Show that the space $(\mathcal{P}, \|\cdot\|)$ of all polynomials on $[0, 1]$ with the supnorm is a normed linear space that is not complete.

(**Hint:** Let $g(t) = \frac{1}{1-t/2}$ for $0 \leq t \leq 1$. Find a sequence of polynomials $\{f_n\}_{n \geq 1}$ in $\mathcal{P}$ that converge to g in supnorm.)

- 3.24 Show that the function $f(v) \equiv \|v\|$ from a normed linear space $(V, \|\cdot\|)$ to $\mathbb{R}_+$ is continuous.

- 3.25 Let $(V, \|\cdot\|)$ be a normed linear space. Let $S = \{v : \|v\| < 1\}$. Show that S is an open set in V .

- 3.26 Show that the space $\mathcal{P}_k$ of all polynomials on $[0, 1]$ of degree $\leq k$ is a Banach space under the supnorm, i.e., under $\|f\| = \sup\{|f(x)|, 0 \leq x \leq 1\}$.

(**Hint:** Let $p_n(x) = \sum_{j=0}^k a_{nj} x^j$ be a sequence of elements in $\mathcal{P}_k$ that converge in supnorm to some $f(\cdot)$. Show that $\{a_{n1}\}_{n \geq 1}$ converges and recursively, $\{a_{ni}\}_{n \geq 1}$ converges for each i .)

- 3.27 Let μ be a finite measure on $[0, 1]$. Verify that $T_\mu(f) \equiv \int f d\mu$ is a bounded linear functional on $C[0, 1]$ and that $\|T_\mu\| = \mu([0, 1])$.

3.28 Let $(V_i, \|\cdot\|_i)$, $i = 1, 2$, be two normed linear spaces over $\mathbb{R}$.

- (a) Let $T : V_1 \rightarrow V_2$ be a linear operator. Show that if for some x_0 , $\|Tx - Tx_0\| \rightarrow 0$ as $x \rightarrow x_0$, then T is continuous on V_1 and hence bounded.
- (b) Show that if $(V_2, \|\cdot\|_2)$ is complete, then $B(V_1, V_2) \equiv \{T \mid T : V_1 \rightarrow V_2, T \text{ linear and bounded}\}$ is complete under the operator norm defined in (3.1).

In the following, H will denote a real Hilbert space.

- 3.29 (a) Use the Cauchy-Schwarz inequality to show that the function $f(x, y) = \langle x, y \rangle$ is continuous from $H \times H \rightarrow \mathbb{R}$.
- (b) (*Parallelogram law*). Show that in an innerproduct space $(V, \langle \cdot, \cdot \rangle)$, for any $x, y \in V$

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

where $\|x\|^2 = \langle x, x \rangle$.

- 3.30 (a) Let $\{Q_n(x)\}_{n \geq 0}$ be defined on $[0, 2\pi]$ by

$$Q_n(x) = c_n \left(\frac{1 + \cos x}{2} \right)^n$$

where c_n is such that

$$\frac{1}{2\pi} \int_0^{2\pi} Q_n(x) dx = 1.$$

Clearly, $Q_n(\cdot) \geq 0$.

- (i) Verify that for each $\delta > 0$,

$$\sup\{Q_n(x) : \delta \leq x \leq 2\pi - \delta\} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

- (ii) Use this to show that if $f \in C[0, 2\pi]$ and if

$$P_n(t) \equiv \frac{1}{2\pi} \int_0^{2\pi} Q_n(t-s)f(s)ds, \quad n \geq 0, \quad (4.1)$$

then $P_n \rightarrow f$ uniformly on $[0, 2\pi]$.

- (b) Use this to give a proof of the completeness of the class C of trigonometric functions.
- (c) Show that if $f \in L^1([0, 2\pi])$, then $P_n(\cdot)$ converges to f in $L^1([0, 2\pi])$.

- (d) Let $\{\mu_n(\cdot)\}_{n \geq 1}$ be a sequence of probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that for each $\delta > 0$, $\mu_n(\{x : |x| \geq \delta\}) \rightarrow 0$ as $n \rightarrow \infty$. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be Borel measurable. Let $f_n(x) \equiv \int f(x-y)\mu_n(dy)$, $n \geq 1$. Assuming that $f_n(\cdot)$ is well defined and Borel measurable, show that
- (i) $f(\cdot)$ continuous at x_0 and bounded $\Rightarrow f_n(x_0) \rightarrow f(x_0)$.
 - (ii) $f(\cdot)$ uniformly continuous and bounded on $\mathbb{R} \Rightarrow f_n \rightarrow f$ uniformly on $\mathbb{R}$.
 - (iii) $f \in L^p(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$, $0 < p < \infty$, $m(\cdot) = \text{Lebesgue measure}$
 $\Rightarrow \int |f_n - f|^p dm \rightarrow 0$.
 - (iv) Show that (iii) $\Rightarrow$ (c).

3.31 (*Gram-Schmidt procedure*). Let $B \equiv \{b_n : n \in \mathbb{N}\}$ be a collection of nonzero vector in H . Set

$$\begin{aligned} e_1 &= \frac{b_1}{\|b_1\|} \\ \tilde{e}_2 &= b_2 - \langle b_2, e_1 \rangle e_1, \\ e_2 &= \frac{\tilde{e}_2}{\|\tilde{e}_2\|} \quad (\text{provided } \|\tilde{e}_2\| \neq 0), \text{ and so on.} \end{aligned}$$

If $\|\tilde{e}_n\| = 0$ for some $n \in \mathbb{N}$, then delete b_n . Let $E \equiv \{e_j : 1 \leq j < k\}$, $k \leq \infty$, be the collection of vectors reached this way.

- (a) Show that E is an orthonormal system.
- (b) Let H_B denote the closed linear subspace generated by B , i.e.,

$$H_B \equiv \left\{ x : x \in H, \text{ there exists } x_n \text{ of the form } \sum_{j=1}^n a_j b_j, \right. \\ \left. a_j \in \mathbb{R}, \text{ such that } \|x_n - x\| \rightarrow 0 \right\}.$$

Show that H_B is a Hilbert space and E is a basis for H_B .

3.32 Let $H = L^2(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$, where μ is a probability measure. Let $B \equiv \{1, x, x^2, \dots\}$. Assume that $\int |x|^k d\mu < \infty$ for all $k \in \mathbb{N}$. Apply the Gram-Schmidt procedure in Problem 3.31 to the set B for the following cases and evaluate e_1, e_2, e_3 .

- (a) $\mu = \text{Uniform } [0, 1] \text{ distribution.}$
- (b) $\mu = \text{standard normal distribution.}$
- (c) $\mu = \text{Exponential (1) distribution.}$

The orthonormal basis E obtained this way is called *Orthogonal Polynomials* w.r.t. the *given measure*. (See Szego (1939).)

- 3.33 Let $B \subset H$ be an orthonormal system. Show that for any x in H , $\{b : \langle x, b \rangle \neq 0\}$ is at most countable.

(**Hint:** Show first that if $\{y_\alpha : \alpha \in I\}$ is a collection of nonnegative real numbers such that for some $C < \infty$, $\sum_{\alpha \in F} y_\alpha \leq C$ for all $F \subset I$, F finite, then $\{\alpha : y_\alpha > 0\}$ is at most countable and apply this to the Bessel inequality.)

- 3.34 Let $B \subset H$. Define $B^\perp \equiv \{x : x \in H, \langle x, b \rangle = 0, \text{ for all } b \in B\}$. Show that $B^\perp$ is a closed subspace of H .

- 3.35 Let $B \subset H$ be a closed subspace of H .

- (a) Using the fact that every Hilbert space admits an orthonormal basis, show that every x in H can be uniquely decomposed as

$$x = y + z \quad (4.2)$$

where $y \in B$ and $z \in B^\perp$ and $\|x\|^2 = \|y\|^2 + \|z\|^2$.

- (b) Let $P_B : H \rightarrow B$ be defined by $P_B x = y$ where x admits the decomposition in (4.2) above. Verify that P_B is a bounded linear operator from H to B and is of norm 1 if B has at least one nonzero vector. (The operator P_B is called the *projection* onto B .)

- (c) Verify that $P_B(P_B x) = P_B x$ for all x in H .

- 3.36 Let H be separable and $\{x_n\}_{n \geq 1} \subset H$ be such that $\{\|x_n\|_{n \geq 1}\}$ is bounded by some $C < \infty$. Show that there exist a subsequence $\{x_{n_j}\}_{j \geq 1} \subset \{x_n\}_{n \geq 1}$ and an x_0 in H , such that for every y in H ,

$$\langle x_{n_j}, y \rangle \rightarrow \langle x_0, y \rangle.$$

(**Hint:** Fix an orthonormal basis $B \equiv \{b_n\}_{n \geq 1} \subset H$. Let $a_{ni} = \langle x_n, b_i \rangle$, $n \geq 1, i \geq 1$. Using $\sum_{i=1}^{\infty} a_{ni}^2 \leq C$ for all n and the Bolzano-Wierstrass property, show that

- (a) there exists $\{n_j\}_{j \geq 1}$ such that $\lim_{j \rightarrow \infty} a_{n_j i} = a_i$ exists for all $i \geq 1$,

$$\sum_{i=1}^{\infty} a_i^2 < \infty,$$

- (b) $\lim_{n \rightarrow \infty} \sum_{i=1}^n a_i b_i \equiv x_0$ exists in H ,

- (c) $\langle x_{n_j}, y \rangle \rightarrow \langle x_0, y \rangle$ for all y in H .)

- 3.37 Let $(V, \langle \cdot, \cdot \rangle)$ be an innerproduct space. Verify that $\langle \cdot, \cdot \rangle$ is *bilinear*, i.e., for $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}$, x_1, x_2, y_1, y_2 in V ,

$$\begin{aligned} \langle \alpha_1 x_1 + \alpha_2 x_2, \beta_1 y_1 + \beta_2 y_2 \rangle &= \alpha_1 \beta_1 \langle x_1, y_1 \rangle + \alpha_1 \beta_2 \langle x_1, y_2 \rangle \\ &\quad + \alpha_2 \beta_1 \langle x_2, y_1 \rangle + \alpha_2 \beta_2 \langle x_2, y_2 \rangle. \end{aligned}$$

State and prove an extension to more than two vectors.

3.38 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Suppose that there exists an at most countable family $\mathcal{A} \equiv \{A_j\}_{j \geq 1} \subset \mathcal{F}$ such that $\mathcal{F} = \sigma\langle \mathcal{A} \rangle$ and $\mu(A_j) > 0$ for each $j \geq 1$. Then show that for $0 < p < \infty$, $L^p(\Omega, \mathcal{F}, \mu)$ is separable.

(**Hint:** Show first that for any $A \in \mathcal{F}$ with $\mu(A) < \infty$, and $\epsilon > 0$, there exists a countable subcollection $\mathcal{A}_1$ of $\mathcal{A}$ such that $\mu(A \triangle B) < \epsilon$ where $B = \{\cup A_j : A_j \in \mathcal{A}_1\}$. Now consider the class of functions $\{\sum_{i=1}^n c_i I_{A_i}, n \geq 1, A_i \in \mathcal{A}, c_i \in \mathbb{Q}\}$.)

3.39 Show that B in Example 3.3.1 (c) is a basis for H .

(**Hint:** Using Theorem 2.3.14 prove that the set of all polynomials are dense in H .)

3.40 (*Haar functions*). For x in $\mathbb{R}$ let $h(x) = I_{[0,1/2)}(x) - I_{[1/2,1)}(x)$. Let $h_{00}(x) \equiv I_{[0,1)}(x)$ and for $k \geq 1$, $0 \leq j < 2^{k-1}$, let $h_{kj}(x) \equiv 2^{\frac{k-1}{2}} h(2^{k-1}x - j)$, $0 \leq x < 1$.

(a) Verify that the family $\{h_{kj}(\cdot), k \geq 0, 0 \leq j < 2^{k-1}\}$ is an orthonormal set in $L^2([0, 1], \mathcal{B}([0, 1]), m)$, where $m(\cdot)$ is Lebesgue measure.

(b) Verify that this family is complete by completing the following two proofs:

(i) Show that for indicator function f of dyadic interval of the form $[\frac{k}{2^n}, \frac{\ell}{2^n})$, $k < \ell$, the following identity holds:

$$\int f^2 dm = \frac{\ell - k}{2^n} = \sum_{k,j} \left(\int f h_{kj} dm \right)^2.$$

Now use the fact the linear combinations of such f 's is dense in $L^2[0, 1]$.

(ii) For each $f \in L^2([0, 1], \mathcal{B}([0, 1]), m)$ such that f is orthogonal to the Haar functions, $F(t) \equiv \int_{[0,t]} f dm$, $0 \leq t \leq 1$ is continuous and satisfies $F(\frac{j}{2^n}) = 0$ for all $0 \leq j \leq 2^n$, $n = 1, 2, \dots$ and hence $F \equiv 0$ implying $f = 0$ a.e.

3.41 Let H be a Hilbert space over $\mathbb{R}$ and M be a closed subspace of H . Let $v_0 \in H$. Show that

$$\min\{\|v - v_0\| : v \in M\} = \max\{\langle v_0, u \rangle, u \in M^\perp, \|u\| = 1\},$$

where $M^\perp$ is the orthogonal complement of M , i.e., $M^\perp \equiv \{u : \langle v, u \rangle = 0 \text{ for all } v \in M\}$.

(**Hint:** Use Problem 3.35 (a).)

3.42 Let B be an orthonormal set in a Hilbert space H .

- (a) (i) Show that for any x in H and any finite set $\{b_i : 1 \leq i \leq k\} \subset B$, $k < \infty$,

$$\sum_{i=1}^k \langle x, b_i \rangle^2 \leq \|x\|^2.$$

- (ii) Conclude that for all x in H , $B_x \equiv \{b : \langle x, b \rangle \neq 0, b \in B\}$ is at most countable.

- (b) Show that the following are equivalent:

- (i) B is complete, i.e., $x \in H$, $\langle x, b \rangle = 0$ for all $b \in B \Rightarrow x = 0$.
(ii) For all $x \in H$, there exists an at most countable set $B_x \equiv \{b_i : i \geq 1\}$ such that $\|x\|^2 = \sum_{i=1}^{\infty} \langle x, b_i \rangle^2$.
(iii) For all $x \in H$, $\epsilon > 0$, there exists a finite set $\{b_1, b_2, \dots, b_k\} \subset B$ such that

$$\left\| x - \sum_{i=1}^k \langle x, b_i \rangle b_i \right\| < \epsilon.$$

- (iv) If $B \subset B^1$, B^1 an orthonormal set in $H \Rightarrow B = B^1$.

3.43 Extend Theorem 3.3.3 to any Hilbert space assuming that the axiom of choice holds.

(Hint: Using the axiom of choice or its equivalent, the Hausdorff maximality principle, it can be shown that every Hilbert space H admits an orthonormal basis B (see Rudin (1987)). Now let T be a bounded linear functional from H to $\mathbb{R}$. Let $f(b) \equiv T(b)$ for b in B . Verify that $\sum_{i=1}^k |f(b_i)|^2 \leq \|T\|^2$ for all finite collection $\{b_i : 1 \leq i \leq k\} \subset B$. Conclude that $D \equiv \{b : f(b) \neq 0\}$ is countable. Let $x_0 \equiv \sum_{b \in D} f(b)b$. Now use the proof of Theorem 3.3.3 to show that $T(x) \equiv \langle x, x_0 \rangle$ for all x in H .)

3.44 Let $(V, \|\cdot\|)$ be a normed linear space. Let $\{T_n\}_{n \geq 1}$ and T be bounded linear operators from V to V . The sequence $\{T_n\}_{n \geq 1}$ is said to converge

- (a) *weakly* to T if for each w in V^* , the dual of V , and each v in V ,

$$w(T_n(v)) \rightarrow w(T(v)),$$

- (b) *strongly* if for each v in V , $\|T_n v - T v\| \rightarrow 0$,

- (c) *uniformly* if $\sup\{\|T_n v - T v\| : \|v\| \leq 1\} \rightarrow 0$.

Let $V_p = L^p(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_L)$, $1 \leq p \leq \infty$. Let $\{h_n\}_{n \geq 1} \subset \mathbb{R}$ be such that $h_n \neq 0$, $h_n \rightarrow 0$, as $n \rightarrow \infty$. Let $(T_n f)(\cdot) \equiv f(\cdot + h_n)$, $Tf(\cdot) \equiv f(\cdot)$. Verify that

- (i) $\{T_n\}_{n \geq 1}$ and T are bounded linear operators on V_p , $1 \leq p \leq \infty$,
- (ii) for $1 \leq p < \infty$, $\{T_n\}_{n \geq 1}$ converges to T weakly,
- (iii) for $1 \leq p < \infty$, $\{T_n\}$ converges to T strongly,
- (iv) for $1 \leq p < \infty$, $\{T_n\}$ does not converge to T uniformly by showing that for all n , $\|T_n - T\| = 1$,
- (v) for $p = \infty$, show that T_n does not converge weakly to T .